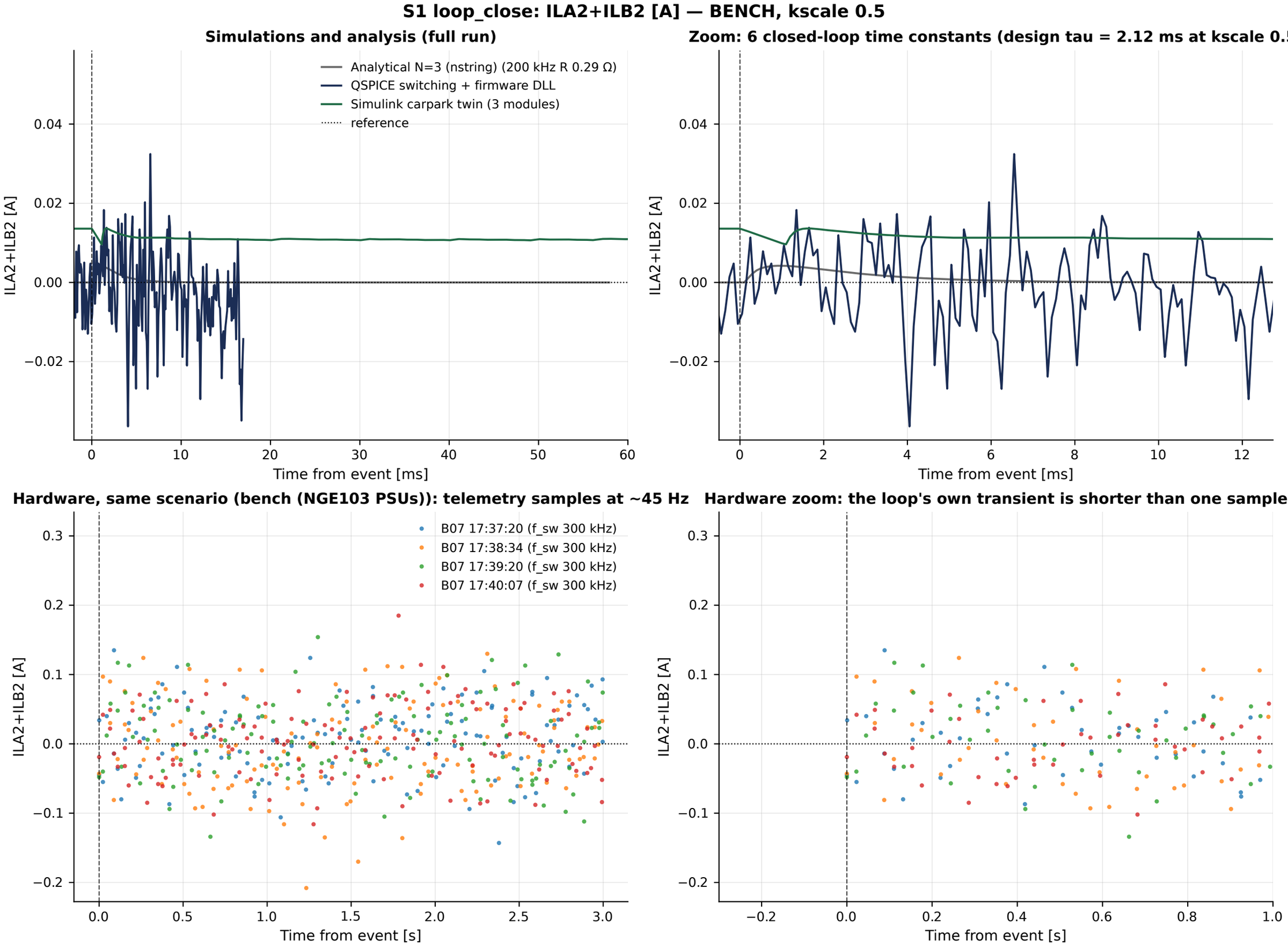
S1 loop_close: ILA2+ILB2 [A] — BENCH, kscale 0.5
Simulations and analysis (full run)
Zoom: 6 closed-loop time constants (design tau = 2.12 ms at kscale 0.5
Analytical N=3 (nstring) (200 kHz R 0.29 Ω)
QSPICE switching + firmware DLL
Simulink carpark twin (3 modules)
reference
ILA2+ILB2 [A]
Time from event [ms]
Hardware, same scenario (bench (NGE103 PSUs)): telemetry samples at ~45 Hz
Hardware zoom: the loop's own transient is shorter than one sample
B07 17:37:20 (f_sw 300 kHz)
B07 17:38:34 (f_sw 300 kHz)
B07 17:39:20 (f_sw 300 kHz)
B07 17:40:07 (f_sw 300 kHz)
Time from event [s]